ATM TRANSACTION SYSTEM

Entity-Relationship Diagram Project

Submitted By:
Shagun Srivastava (2410030277)
Simarpreet Kaur (2410030200)
Prachi Rautela(2410030248)

Course: B.TechCSE (DS&BD)
Subject: DatabaseManagement System(DBMS)
Submitted To: Mr.Atul Kumar Singh
University: IILMUniversity, Greater Noida
Submission Date: 23/02/2026

Declaration

We hereby declare that this project titled 'ATM Transaction System - ER Diagram' is an original work carried out by us under the guidance of Mr. Atul Kumar Singh. This project has not been submitted previously for any academic evaluation.

Shagun Srivastava (2410030277)

Simarpreet Kaur (2410030200)

Prachi Rautela (2410030248)

1. Introducion

The ATM Transaction System is designed to manage records of banks, customer accounts, ATM cards, ATM machines, and the financial transactions performed. The system ensures efficient tracking of account balances, card statuses, and real-time transaction logs across multiple ATM locations. The ER diagram provides a conceptual representation of the database structure.

2. Entities and Attributes

BANK

- Bank_ID (Primary Key)
- IFSC_code
- Bank_Name
- Head_office

BANK_Account

- Acc_No. (Primary Key)
- Acc_Type
- Balance

CUSTOMER

- Customer_ID (Primary Key)
- Name
- Phone_number
- Address
- E-mail

ATM_CARD

- Card_no. (Primary Key)
- PIN
- Expiry_Date
- Card_Type
- Card_Status

ATM MACHINE

- ATM_ID (Primary Key)
- Location
- STATUS
- Cash_Avail
- ATM_Status
- Bank_name

TRANSACTION (Weak Entity/Dependent on Card & ATM)

- Trans_ID (Primary Key / Partial Key)
- Trans_type
- Date_time
- Amount
- Status

3. Relationships

1.Has (BANK – BANK_Account): One bank has multiple bank accounts (1:N).

2.Owns (BANK – ATM MACHINE): One bank owns multiple ATM machines (1:N).

3.Owns (CUSTOMER – ATM_CARD): One customer can own multiple ATM cards (1:N).

4. Linked_To (BANK_Account – ATM_CARD): One bank account can be linked to multiple ATM cards (1:N).

5. Performs (ATM_CARD – TRANSACTION): One ATM card can perform multiple transactions (1:N).

6. Done_at (ATM MACHINE – TRANSACTION): One ATM machine processes multiple transactions (1:N).

4. Relational Schema

BANK(Bank_ID PK, IFSC_code, Bank_Name, Head_office)

BANK_ACCOUNT(Acc_No PK, Acc_Type, Balance, Bank_ID FK)

CUSTOMER(Customer_ID PK, Name, Phone_number, Address, E-mail)

ATM_CARD(Card_no PK, PIN, Expiry_Date, Card_Type, Card_Status, Customer_ID FK, Acc_No FK)

ATM_MACHINE(ATM_ID PK, Location, STATUS, Cash_Avail, ATM_Status, Bank_name, Bank_ID FK)

TRANSACTION(Trans_ID PK, Trans_type, Date_time, Amount, Status, Card_no FK, ATM_ID FK)

5. SQL Table Creation Queries

```
CREATE TABLE BANK (
    Bank_ID INT PRIMARY KEY,
    IFSC_code VARCHAR(20),
    Bank_Name VARCHAR(100),
    Head_office VARCHAR(150)
);

CREATE TABLE BANK_ACCOUNT (
    Acc_No INT PRIMARY KEY,
    Acc_Type VARCHAR(50),
    Balance DECIMAL(15,2),
    Bank_ID INT,
    FOREIGN KEY (Bank_ID) REFERENCES BANK(Bank_ID)
);
```

```
CREATE TABLE CUSTOMER (
    Customer_ID INT PRIMARY KEY,
    Name VARCHAR(100),
    Phone_number VARCHAR(15),
    Address VARCHAR(255),
    Email VARCHAR(100)
);
CREATE TABLE ATM_CARD (
    Card_no INT PRIMARY KEY,
    PIN INT,
    Expiry_Date DATE,
    Card_Type VARCHAR(50),
    Card_Status VARCHAR(50),
    Customer_ID INT,
    Acc_No INT,
    FOREIGN KEY (Customer_ID) REFERENCES CUSTOMER(Customer_ID),
    FOREIGN KEY (Acc_No) REFERENCES BANK_ACCOUNT(Acc_No)
);

CREATE TABLE ATM_MACHINE (
    ATM_ID INT PRIMARY KEY,
    Location VARCHAR(255),
    STATUS VARCHAR(50),
    Cash_Avail DECIMAL(15,2),
    ATM_Status VARCHAR(50),
    Bank_name VARCHAR(100),
    Bank_ID INT,
    FOREIGN KEY (Bank_ID) REFERENCES BANK(Bank_ID)
);

CREATE TABLE TRANSACTION (
    Trans_ID INT PRIMARY KEY,
    Trans_type VARCHAR(50),
    Date_time DATETIME,
    Amount DECIMAL(15,2),
    Status VARCHAR(50),
    Card_no INT,
    ATM_ID INT,
    FOREIGN KEY (Card_no) REFERENCES ATM_CARD(Card_no),
    FOREIGN KEY (ATM_ID) REFERENCES ATM_MACHINE(ATM_ID)
);
```

#Output Tables:

```
MariaDB [DBMS]> show tables;
+------------------+
| Tables_in_dbms |
+------------------+
| atm_card         |
| atm_machine      |
| bank             |
| bank_account     |
| customer         |
| transaction      |
+------------------+
6 rows in set (0.001 sec)
```

DESC bank;

DESC bank_account;

DESC customer;

DESC atm_card;

DESC atm_machine;

DESC transaction;

Showing table of every entity

```
MariaDB [DBMS]> DESC bank;
+-------------+--------------+------+-----+---------+-------+
| Field       | Type         | Null | Key | Default | Extra |
+-------------+--------------+------+-----+---------+-------+
| Bank_ID     | int(11)      | NO   | PRI | NULL    |       |
| IFSC_code   | varchar(20)  | YES  |     | NULL    |       |
| Bank_Name   | varchar(100) | YES  |     | NULL    |       |
| Head_office | varchar(150) | YES  |     | NULL    |       |
+-------------+--------------+------+-----+---------+-------+
4 rows in set (0.019 sec)
```

```
MariaDB [DBMS]> DESC bank_account;
+----------+---------------+------+-----+---------+-------+
| Field    | Type          | Null | Key | Default | Extra |
+----------+---------------+------+-----+---------+-------+
| Acc_No   | int(11)       | NO   | PRI | NULL    |       |
| Acc_Type | varchar(50)   | YES  |     | NULL    |       |
| Balance  | decimal(15,2) | YES  |     | NULL    |       |
| Bank_ID  | int(11)       | YES  | MUL | NULL    |       |
+----------+---------------+------+-----+---------+-------+
4 rows in set (0.030 sec)
```

```
MariaDB [DBMS]> DESC customer;
+--------------+--------------+------+-----+---------+-------+
| Field        | Type         | Null | Key | Default | Extra |
+--------------+--------------+------+-----+---------+-------+
| Customer_ID  | int(11)      | NO   | PRI | NULL    |       |
| Name         | varchar(100) | YES  |     | NULL    |       |
| Phone_number | varchar(15)  | YES  |     | NULL    |       |
| Address      | varchar(255) | YES  |     | NULL    |       |
| Email        | varchar(100) | YES  |     | NULL    |       |
+--------------+--------------+------+-----+---------+-------+
5 rows in set (0.027 sec)
```

```
ariaDB [DBMS]> DESC atm_card;
-------------+-------------+------+-----+---------+-------
Field        | Type        | Null | Key | Default | Extra
-------------+-------------+------+-----+---------+-------
Card_no      | int(11)     | NO   | PRI | NULL    |
PIN          | int(11)     | YES  |     | NULL    |
Expiry_Date  | date        | YES  |     | NULL    |
Card_Type    | varchar(50) | YES  |     | NULL    |
Card_Status  | varchar(50) | YES  |     | NULL    |
Customer_ID  | int(11)     | YES  | MUL | NULL    |
Acc_No       | int(11)     | YES  | MUL | NULL    |
-------------+-------------+------+-----+---------+-------
rows in set (0.029 sec)
```

```
ariaDB [DBMS]> DESC atm_machine;
+-----------+---------------+------+-----+---------+-------+
| Field     | Type          | Null | Key | Default | Extra |
+-----------+---------------+------+-----+---------+-------+
| ATM_ID    | int(11)       | NO   | PRI | NULL    |       |
| Location  | varchar(255)  | YES  |     | NULL    |       |
| STATUS    | varchar(50)   | YES  |     | NULL    |       |
| Cash_Avail | decimal(15,2) | YES  |     | NULL    |       |
| ATM_Status | varchar(50)   | YES  |     | NULL    |       |
| Bank_name | varchar(100)  | YES  |     | NULL    |       |
| Bank_ID   | int(11)       | YES  | MUL | NULL    |       |
+-----------+---------------+------+-----+---------+-------+
rows in set (0.021 sec)
```

```
ariaDB [DBMS]> DESC transaction;
+-----------+---------------+------+-----+---------+-------+
| Field     | Type          | Null | Key | Default | Extra |
+-----------+---------------+------+-----+---------+-------+
| Trans_ID  | int(11)       | NO   | PRI | NULL    |       |
| Trans_type | varchar(50)   | YES  |     | NULL    |       |
| Date_time | datetime      | YES  |     | NULL    |       |
| Amount    | decimal(15,2) | YES  |     | NULL    |       |
| Status    | varchar(50)   | YES  |     | NULL    |       |
| Card_no   | int(11)       | YES  | MUL | NULL    |       |
| ATM_ID    | int(11)       | YES  | MUL | NULL    |       |
+-----------+---------------+------+-----+---------+-------+
rows in set (0.017 sec)
```

6. Conclusion

The ATM Transaction System ER model effectively represents the database structure required for managing banking operations, customer data, and secure machine transactions. It minimizes redundancy and maintains data consistency across bank branches and ATMs.